

Knife Crime in London

Exploratory Data Analysis of Metropolitan Police monthly crime data, August 2022 – July 2026

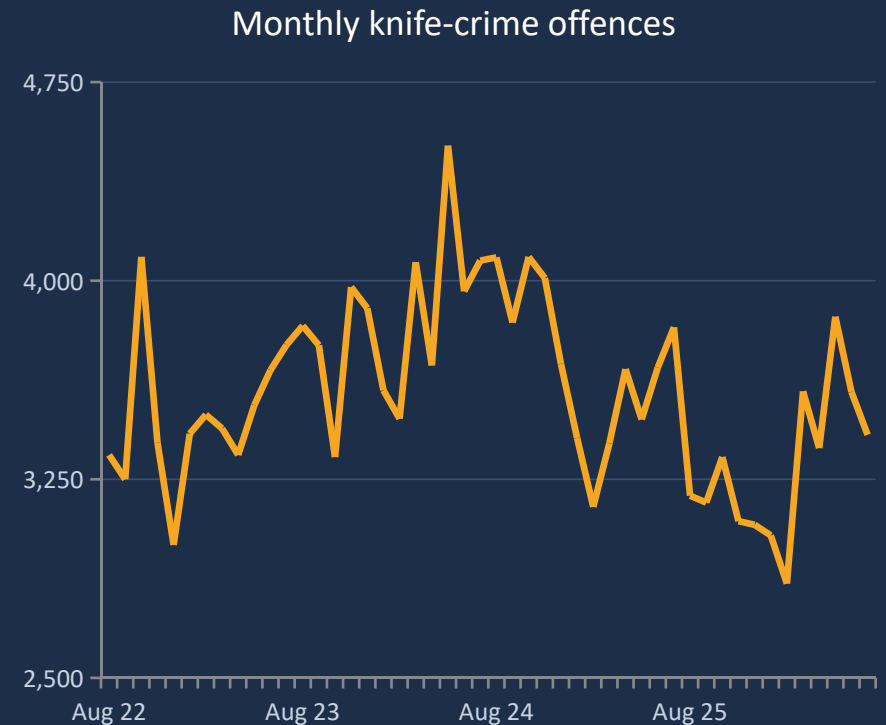
INDUSTRY Public Safety & Law Enforcement

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COURSE AIML

BATCH CODE ANP-D7444



171,727

offence count across 48 months

Industry Overview



SELECTED INDUSTRY

Public Safety & Law Enforcement

Police services prevent, detect and investigate crime and keep communities safe. In London, the Metropolitan Police Service publishes monthly offence data by borough and Safer Neighbourhood Team.

Why data analysis matters here



Spot patterns early

Trends, seasonality and hotspots show up in the numbers before they are obvious on the ground.



Deploy resources wisely

Evidence on where and when offences occur guides patrols and prevention work.



Measure what works

Year-on-year comparisons show whether interventions are reducing harm.



Build public trust

Open, transparent data lets communities and policy makers see the same facts.

Problem Statement



The problem

Knife crime is a high-harm offence that remains a persistent challenge across London. Without structured analysis of the monthly records, it is hard to see where it concentrates, when it peaks and whether it is declining.



Why it matters

Knife offences can cause serious injury and loss of life, fuel fear in communities and place heavy demand on police and emergency services. Better evidence supports earlier, better-targeted action.



Who is affected

- Victims and their families, including young people (the data tracks victims aged 1–24)
- Residents and local communities
- Police and Safer Neighbourhood Teams
- Councils, policy makers and health services

Core question: Where, when and in what form does knife crime occur across London, and is it falling?

Project Objectives

5

clear objectives

Each objective maps to analysis questions answered in the notebook.

1

Clean and prepare

Fix duplicates, missing values and inconsistent labels so the data is reliable.

2

Measure change over time

Track knife-crime offences by month and by financial year.

3

Locate high-crime areas

Rank boroughs, Safer Neighbourhood Teams and area-month hotspots.

4

Compare crime types

Break down subtypes and compare knife crime with injury.

5

Reveal seasonality and variation

Find which months and areas swing the most, to support planning.

Proposed Solution & Analysis Questions

Proposed solution

A reproducible Python pipeline in Jupyter that loads, cleans and analyses the monthly knife-crime data, then answers key questions with tables and charts.

How Python helps

- Pandas cleans and aggregates 60,000+ records in seconds
- NumPy supports the numeric calculations behind the statistics
- Matplotlib turns results into clear, repeatable charts



Load



Clean



Analyse



Visualise

Analysis questions

Q1

How do knife-crime offences change over time?

Q2

Which financial year recorded the most offences?

Q3

Which boroughs have the highest offences?

Q4

How is knife crime split by subtype?

Q5

Which months show a seasonal peak?

Q6

Which areas and months are hotspots?

The notebook answers 10 questions; six are highlighted here.

Dataset Information

DATASET

MonthlyCrimeDashboard_KnifeCrimeData.csv

SOURCE

Metropolitan Police Service – Monthly Crime Dashboard (London); refreshed 18 Aug 2026

FORMAT

CSV (comma-separated values)

60,197

rows (60,100 after cleaning)

11

columns

48

months, Aug 2022 – Jul 2026

Monthly counts of knife-crime offences and positive outcomes for London boroughs and Safer Neighbourhood Teams, by crime subtype and financial year.

Column	Description	Example
Month_Year	Month of the record	2025-02-01
Area Type	Borough or Safer Neighbourhood Team	Borough
Borough_SNT	Borough or team name	Lambeth Brixton Windrush
Area Name	Area / ward name	Brixton Windrush
Area Code	Official area code	E05014098
Crime Type	Always Knife Crime	Knife Crime
Crime Subtype	5 subtypes, e.g. with injury	Knife Crime with Injury
Measure	Offences or Positive Outcomes	Offences
Financial Year	fy22-23 to fy26-27	fy24-25
Count	Number of offences in the row	2
Refresh Date	Date the data was refreshed	2026-08-18

Data Collection & Understanding

Collection

Monthly knife-crime data published by the Metropolitan Police Service, loaded with `pd.read_csv()`.

Structure

60,197 rows × 11 columns covering Aug 2022 – Jul 2026.

Data types

10 text columns and 1 integer (Count); dates arrive as text.

Missing values

243 blank Borough_SNT and Area Name; 276 blank Area Code.

Duplicates

97 exact duplicate rows.

```
In [7]: clean_df.info()

<class 'pandas.DataFrame'>
RangeIndex: 60197 entries, 0 to 60196
Data columns (total 11 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Month_Year      60197 non-null  str    
1   Area Type       60197 non-null  str    
2   Borough_SNT     59954 non-null  str    
3   Area Name       59954 non-null  str    
4   Area Code       59921 non-null  str    
5   Crime Type      60197 non-null  str    
6   Crime Subtype   60197 non-null  str    
7   Measure         60197 non-null  str    
8   Financial Year   60197 non-null  str    
9   Count           60197 non-null  int64  
10  Refresh Date    60197 non-null  str    
dtypes: int64(1), str(10)
memory usage: 12.6 MB
```

`df.info()`: shape, types, non-null counts

```
In [8]: clean_df.isnull().sum()

Month_Year      0
Area Type       0
Borough_SNT     243
Area Name       243
Area Code       276
Crime Type      0
Crime Subtype   0
Measure         0
Financial Year   0
Count           0
Refresh Date     0
dtype: int64
```

```
In [9]: clean_df.duplicated().sum()

np.int64(97)
```

`isnull().sum()` and `duplicated().sum()`

60,197 rows loaded

243 blank area names

97 duplicate rows

Data Cleaning & Transformation

- 1 Remove duplicates**
97 exact duplicate rows dropped: 60,197 → 60,100.
- 2 Correct data types**
Month_Year converted from text to datetime.
- 3 Handle missing values**
Blank Borough_SNT, Area Name and Area Code set to Unknown.
- 4 Fix inconsistent labels**
N/K and 'Geography Not Known' standardised; code -1 set to Unknown.
- 5 Filter data**
Measure = Offences only: 48,927 records (Positive Outcomes excluded).
- 6 Create new columns**
Month and Month_Name extracted from Month_Year.

```
In [16]: # Remove exact duplicate rows from clean_df
clean_df = clean_df.drop_duplicates()

print("Original rows:", len(raw_df))
print("Rows after removing duplicates:", len(clean_df))
print("Duplicates remaining:", clean_df.duplicated().sum())

Original rows: 60197
Rows after removing duplicates: 60100
Duplicates remaining: 0

In [17]: # Convert Month_Year to datetime
clean_df['Month_Year'] = pd.to_datetime(clean_df['Month_Year'])

print(clean_df['Month_Year'].dtype)
print(clean_df['Month_Year'].head())

datetime64[us]
0    2025-02-01
1    2025-12-01
2    2024-01-01
3    2024-12-01
4    2023-02-01
Name: Month_Year, dtype: datetime64[us]
```

Remove duplicates, convert Month_Year to datetime

```
In [18]: clean_df['Borough_SNT'] = clean_df['Borough_SNT'].fillna('Unknown')
clean_df['Area Name'] = clean_df['Area Name'].fillna('Unknown')
clean_df['Area Code'] = clean_df['Area Code'].fillna('Unknown')
clean_df = clean_df.replace(to_replace=r'(?i)^N/K.*$',
                           value='Unknown', regex=True)
clean_df['Area Type'] = clean_df['Area Type'].replace(
    {'Borough (Geography Not Known)': 'Borough',
     'Safer Neighbourhood Teams (Geography Not Known)':
      'Safer Neighbourhood Teams'})

Missing Values after cleaning:
Month_Year      0
Area Type       0
Borough_SNT     0
Area Name       0
Area Code       0
Crime Type      0
Crime Subtype   0
Measure         0
Financial Year   0
Count           0
Refresh Date    0
dtype: int64
```

Fill and standardise unknown geography; 0 missing after

Result: 0 missing values across all 11 columns. 4 duplicate rows appeared after labels were standardised; they were flagged for follow-up, not removed.

Exploratory Data Analysis

171,727

Total offence count

3,578

Average per month

4,510

Peak: May 2024

2,856

Low: Feb 2026

Libraries used

Pandas: groupby, agg, pct_change, idxmax

NumPy: numeric backend for mean and std

Matplotlib: 10 charts, one per question

Key findings

- 24 of 48 months were above the monthly average.
- FY25-26 offences fell 12.9% from FY24-25, and 9.6% below FY23-24.
- Croydon varies most month to month (std 25.1, range 50–165).

	Financial Year	Total_Offences	Records	Percentage_Change
0	fy22-23	27430	7896	NaN
1	fy23-24	44219	12264	61.206708
2	fy24-25	45880	12595	3.756304
3	fy25-26	39968	12059	-12.885789
4	fy26-27	14230	4113	-64.396517

Financial year summary

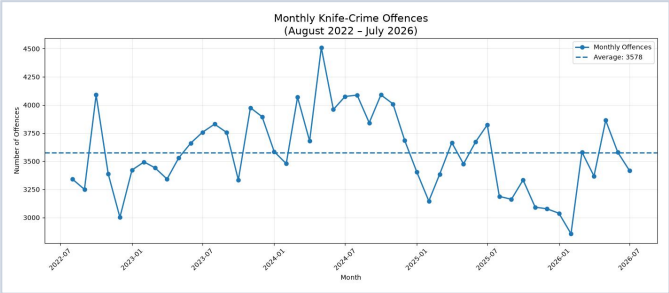
FY22-23 and FY26-27 are partial years, so comparisons use the three complete years.

	Mean	Std_Dev	Min	Max	Months
Borough_SNT					
Croydon	90.916667	25.092877	50	165	48
Ealing	68.895833	20.674004	28	141	48
Southwark	91.333333	20.562653	50	135	48
Lambeth	94.437500	19.877183	55	147	48
Brent	75.958333	19.678218	35	118	48
Westminster	92.916667	19.502137	55	141	48
Newham	94.979167	18.603923	61	130	48
Haringey	73.750000	17.715842	46	120	48
Tower Hamlets	75.458333	16.314669	39	111	48
Wandsworth	43.729167	15.546416	24	96	48

Top 10 areas by monthly std dev

Croydon, Ealing and Southwark swing the most from month to month.

Data Visualization

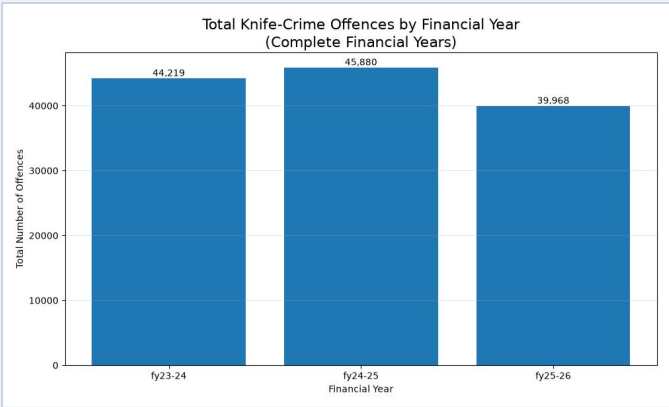


Monthly Offences (line chart)

Purpose: Track the trend over 48 months.

Shows: Monthly totals against the 3,578 average line.

Insight: Flat overall (+2.3%); peak 4,510 in May 2024, low 2,856 in Feb 2026.

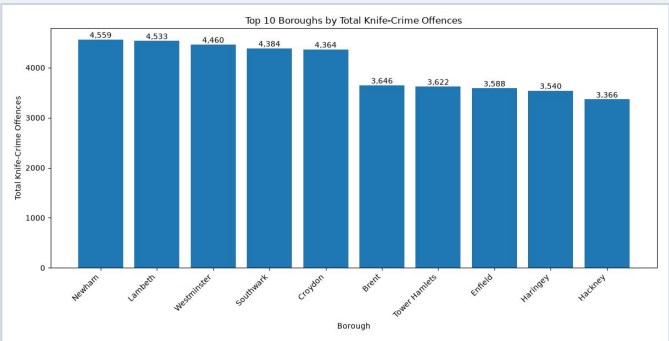


Offences by Financial Year (bar)

Purpose: Compare the complete financial years.

Shows: 44,219, then 45,880, then 39,968 offences.

Insight: FY24-25 was the peak; FY25-26 was 12.9% lower.

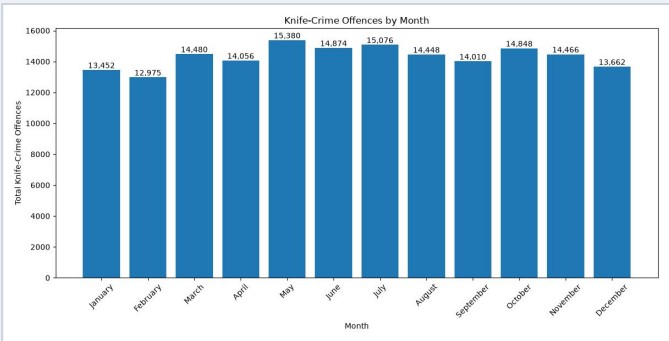


Top 10 Boroughs (bar chart)

Purpose: Find where offences concentrate.

Shows: Newham 4,559 down to Hackney 3,366.

Insight: Newham, Lambeth, Westminster, Southwark and Croydon lead.



Offences by Month of Year (bar)

Purpose: Test for seasonality.

Shows: Totals for each calendar month across 4 years.

Insight: May is highest (15,380); February is lowest (12,975).

Key Insights & Recommendations

+2.3%	Offences were flat: 3,342 in Aug 2022 and 3,418 in Jul 2026.
-12.9%	FY25-26 fell from the FY24-25 peak of 45,880 to 39,968.
May	Highest month of the year (15,380); February is lowest (12,975).
Newham	Top borough with 4,559, ahead of Lambeth, Westminster, Southwark, Croydon.
Croydon	Most volatile area (std 25.1) and top hotspot: 165 in Sep 2023.
24.8%	Knife crime with injury equals 24.8% of the Knife Crime subtype count.



Recommendations

- Focus prevention on the top 5 boroughs and recurring hotspots (Croydon, Lambeth, Southwark).
- Plan extra capacity for May to July.
- Study what drove the FY25-26 fall and repeat it.
- Record geography fully: 2,442 team-level offence records had no area assigned.

Conclusion



What was achieved

Cleaned 60,197 raw records to 60,100, then analysed 48,927 offence records over 48 months with Pandas, NumPy and Matplotlib.



Major findings

Offences were flat overall, peaked in FY24-25 and fell in FY25-26. May is the seasonal high. Newham leads the boroughs and Croydon is the top hotspot.



How it addresses the problem

Shows where and when knife crime concentrates, so prevention and patrol resources can be targeted with evidence.



Future scope

Normalise by population, resolve overlapping Borough/SNT rows, add Seaborn heatmaps and maps, forecast with ML models and study positive outcomes.